Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Problems on formation of PDE by eliminating arbitrary functions

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Problems on formation of PDE by eliminating arbitrary functions

1 Problems on formation of PDE by eliminating arbitrary functions

Study focus: Problems on formation of PDE by eliminating arbitrary functions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000022 · EV0000023

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will solve problems on formation of partial differential equation by eliminating arbitrary functions.

Key details

- In one of the classes we understood what is the procedure to form a PDE by eliminating arbitrary functions.
- And phi becomes phi dash.
- Phi dash means phi dash of x square minus y square and x square become 2x.
- In a similar manner, differentiate z partially with respect to y.
- So we get phi as the first function.
- X plus Y becomes 1 with respect to Y and pi dash of X square minus Y square times minus 2y and this X plus Y is a another function.
- How to eliminate file.
- That is so phi dash becomes phi minus z y divided by two y of x plus y.
- It means you can take this term to right hand side left hand side, it become plus.
- x plus y times u y into phi dash.
- Then you can shift this z sub x y to right hand side so it becomes phi minus z y.
- Write x plus y and two x as it is.
- In place of this phi dash, we substitute phi minus douze by dou y over two y of x plus y.
- By the standard notation, we understood that those z by doux is same as p and this y as it is.

- So we are left out with the x by y of phi minus q because dou z by dou y equals q as per the standard notation.
- If you look at the problem.
- Say phi means phi of x square minus y square.
- This is z over x plus y.
- X by Y times Q.
- You can clearly see that's this is a first order partial differential equation.

Example or application from the lecture

- Let us note on the first problem z equals x plus y phi of x square minus y square.
- So let us take y as the first function and the derivative of x plus y with respect to x is 1 plus x plus y is a second function.
- Let us call this as expression 1.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 05:00–09:59

That's the answer.

Key details

- I hope you understood the concept.
- Z equals x power n times f of y by x.
- Even this function contains a single arbitrary function, namely f.
- We differentiate z partially with respect to x.
- So f is the first function.
- And the derivative of x power n with respect to x is nothing but n times x into x power n minus 1 plus x power n as the first function and the derivative of f of y by x becomes f dash.
- of y by x then y by x becomes minus y by x square with respect to x.
- And f becomes f dash and y by x becomes 1 by x.
- Y is 1 and x is constant.
- Again, this is going to be x power n minus two times y.
- In place of f dash, we put dou z by dou y over x power n minus one.
- So in the next step we can see this is nothing but x power n divided by x.
- And this is going to be if you use law indices for x power n minus 2 and x power n minus 1, you get x power minus 1.
- nx n f minus y into q because those that better y is q as per the standard notation so if you observe this particular step still f is same so how do we eliminate this f.
- Because here we take it is f dash of one plus dou z by dou y.
- It is nothing but this.
- When you take f dash common get one plus p.

Example or application from the lecture

- Let us look at the second problem.

- Okay in a similar manner let us compute to dou z by dou y so for douz by dou y no need to apply product rule because x becomes a constant.
- Let us substitute this into this expression.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 10:00–14:59

And this can be simplified to this particular step.

Key details

- you can just shift.
- one plus q to left hand side y z plus x y p equals one plus p time x z plus x y q so when we simplify this so that leads to this particular step so that is x of y minus z p plus.
- Y of z minus x q equals z of x minus y.
- So you decide what is the order of this linear differential equation because of presence of P and Q.
- With respect to y, we get douz by dou y equals f of x times g dash f i because f of x is constant.
- We proceed for.
- Second order partial derivatives.
- Observe carefully from either of these two.
- You can say either ZXY or it could be ZYX.
- No, this is nothing but mixed partial derivative.
- It is nothing but.
- F dash of x times g dash of y.
- Even the second is also the same.
- dou square z divided by doux dot dou y.
- As you can see, as per the standard notation.
- This is nothing but S.
- by eliminating arbitrary functions.

Example or application from the lecture

- Let us take the product of dou z by doux or dou z by dou y so that leads to this particular expression and this is same as f of x dot g of y dot f dash of x then g dash of y.
- The given problem is f of x dot g of y equals f dash of x dot g dash of y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Problems on formation of PDE by eliminating arbitrary functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__9ce94de93a3b3017.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on formation of PDE by eliminating arbitrary functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__9ce94de93a3b3017.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on formation of PDE by eliminating arbitrary functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__9ce94de93a3b3017.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.
Problems on formation of PDE by eliminating arbitrary functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__9ce94de93a3b3017.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0050. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00011	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0050. Sidecar variants compare...
SRC00012	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0050. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00013	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0050. Sidecar variants compare...

C Revision and retrieval practice